Inequalities Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Dividing by a negative

ORIGINAL PROBLEM

$-3x > 18$

INCORRECT WORKED SOLUTION

$$-3x > 18$$

$$\frac{-3x}{-3} > \frac{18}{-3}$$

$$x > -6$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Distributing a negative

ORIGINAL PROBLEM

$-2(x - 5) \leq 14$

INCORRECT WORKED SOLUTION

$$-2(x - 5) \leq 14$$

$$-2x - 10 \leq 14$$

$$-2x \leq 24$$

$$x \geq -12$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Combining unlike terms

ORIGINAL PROBLEM

$4x + 3 < 2x + 11$

INCORRECT WORKED SOLUTION

$4x + 3 < 2x + 11$

$4x + 3 < 13x$

$3 < 9x$

$x > \frac{1}{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Compound inequality and negatives

ORIGINAL PROBLEM

$-12 < -3x \leq 6$

INCORRECT WORKED SOLUTION

$-12 < -3x \leq 6$

$\frac{-12}{-3} < x \leq \frac{6}{-3}$

$4 < x \leq -2$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. AND versus OR

ORIGINAL PROBLEM

$x > -2$ AND $x < 5$

INCORRECT WORKED SOLUTION

$x > -2$ AND $x < 5$

$x < -2$ OR $x > 5$

Answer: $x < -2$ OR $x > 5$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Open and closed endpoints

ORIGINAL PROBLEM

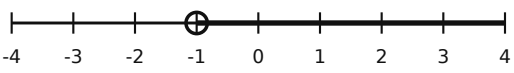
Graph $x \geq -1$

INCORRECT WORKED SOLUTION

$x \geq -1$

Draw an OPEN circle at -1.

Shade right from -1.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Inequalities Error Analysis

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7. Flipping the inequality sign

ORIGINAL PROBLEM

$5 - 2x < 11$

INCORRECT WORKED SOLUTION

$5 - 2x < 11$

$-2x < 6$

$$\frac{-2x}{-2} < \frac{6}{-2}$$

$x < -3$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. OR: forgetting a branch

ORIGINAL PROBLEM

$3x < -6$ OR $x + 1 \geq 5$

INCORRECT WORKED SOLUTION

$3x < -6$ OR $x + 1 \geq 5$

$3x < -6$

$x < -2$

Answer: $x < -2$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Inequalities Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Recognizing no solution

ORIGINAL PROBLEM

$2(x + 3) > 2x + 9$

INCORRECT WORKED SOLUTION

$2(x + 3) > 2x + 9$

$2x + 6 > 2x + 9$

$6 > 9$

Answer: all real numbers

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Writing a real-world inequality

ORIGINAL PROBLEM

Tickets cost $12 each; you have at most $50. How many tickets?

INCORRECT WORKED SOLUTION

$12t \geq 50$

$t \geq \frac{50}{12}$

$t \geq 4.17$

At least 5 tickets

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points